

Yassin Soliman

403-671-2013 | solimanyassin@gmail.com | yassin.app | linkedin.com/in/yassinsoliman | github.com/yassinsolim

SUMMARY

Software Engineering student and AMD AI GPU Software Intern building **ROCm/MIOpen** performance tooling, kernel heuristics, and benchmark automation for AMD Instinct GPUs. Builds across **C/C++**, **Python**, real-time engines, and full-stack systems; targeting **Software Engineer** and **ML/GPU Software** roles.

EDUCATION

University of Calgary, Schulich School of Engineering
Bachelor of Science in Software Engineering

Calgary, AB
Expected April 2028

EXPERIENCE

- Software Engineer Intern, AI GPU Software (AGS) Libraries** May 2025 – Present
AMD | [🔗](#) | [🐞](#) | **C++, Python, ROCm, MIOpen, hipDNN, SLURM, CI/CD** *Calgary, AB*
- Built **MIOpen/hipDNN** performance and benchmark tooling: **SLURM**-driven benchmark orchestration, MCP shape-collection management, and a **Python** performance database viewer, across roughly **10K lines** of Python and shell
 - Prototyped **hipDNN heuristic plugins** (SDPA and convolution regime classifiers, engine-ID probes) and a **ROCm 7.12** MIOpen provider verifier in the public **ROCm** libraries to route deep-learning kernels to faster GPU solvers
 - Reduced representative MIOpen benchmark command sets by **95%** (35K to 1.8K) while preserving **100% solver and config coverage**, cutting **GPU** tuning time
 - Improved deep-learning kernel throughput **~15%** for a production model benchmark suite on **AMD Instinct GPUs** by routing convolution/attention shapes to faster solvers
- Software Lead** Nov. 2024 – Present
Waybionic Club | [🔗](#) | [🐞](#) | **C, C++, Arduino, Next.js, TypeScript, Agile** *Calgary, AB*
- Led development of control systems for a **robotic surgical arm** in **C, C++, and Arduino** for precise, reliable motion control
 - Integrated **sensors, actuators, and real-time embedded algorithms**; built a **Next.js/TypeScript** site to document the project
 - Directed a cross-functional team of 8 engineers through **Agile** sprints and code reviews, improving productivity by **40%**
- Software Engineer** May 2024 – May 2025
Little Footprints | [🔗](#) | [🐞](#) | **JavaScript, HTML, CSS, REST APIs, Agile** *Calgary, AB*
- Built fintech websites using **JavaScript, HTML, and CSS** with **REST API** integration, boosting sales by **30%**
 - Improved frontend performance and **SEO** through web traffic analysis and **SSL** integration, enabling **50% more users**

OPEN SOURCE CONTRIBUTIONS

- Spicetify Marketplace Contributor** | [🐞](#) | **TypeScript, React, Zod, i18next** 2026
- Merged two PRs into the **1.5K+ star** project: [#1209](#) centralized **Zod validation** and cache recovery and passed build, lint, and **CodeQL**; [#1210](#) fixed translations by using Spotify's active client locale

PROJECTS

- Nordschleife Racer** | [🔗](#) | [🐞](#) | **TypeScript, Three.js, WebGL, Supabase** 2026
- Built a **12.7K-line** browser racing engine with custom vehicle physics, raycast suspension, drift handling, and ghost replays
 - Added **real-time multiplayer** over Supabase with a Postgres global leaderboard secured by row-level security; deployed live at yassin.app
- WebStrafe** | [🐞](#) | **TypeScript, Three.js, WebSockets** 2026
- Built a from-scratch **128 Hz** fixed-timestep physics engine with BVH collision, unit-tested movement, real-time multiplayer, and online leaderboard
- PathGuard** | [🔗](#) | [🐞](#) | **Python, YOLOv8, ESP32-CAM, AWS** 2025
- Hackathon team build: **YOLOv8** + ByteTrack pedestrian tracking on ESP32-CAM hardware with an **AWS** serverless backend (Lambda, DynamoDB, SNS)

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, JavaScript, TypeScript, HTML5, CSS
AI / GPU: ROCm, MIOpen, GPU kernels, PyTorch, NumPy, OpenCV, YOLOv8, ByteTrack
Web & Cloud: React, Next.js, Three.js, Node.js, Django, AWS (Lambda, DynamoDB, S3), Docker, Kubernetes
Tools: Git, GitHub, Linux, Shell Scripting, SLURM, CI/CD, Proxmox VE, Jira, Agile